



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2013 - 51



INTELLIGENT MOISTURE TESTER (for palay) Model : LDS-1G

INTELLIGENT LDS-1G GRAIN MOISTURE TESTER (for palay)

Test Report No.	:	2013 - 51
Date of Test	:	June 4 – 24, 2013
Test Valid Until	:	June 24, 2018
Brand	:	INTELLIGENT
Model	:	LDS-1G
Type	:	Portable, Digital, Capacitance
Manufacturer	:	Made in China
Test Requested by	:	MIAITO SALES INTERNATIONAL 49 Columbia St., BLS, Paranaque City, Metro Manila
Machine Classification	:	Commercial

SUMMARY

The INTELLIGENT LDS-1G Grain Moisture Tester was tested for performance. It was tested using fresh palay at different moisture contents. At these different moisture content levels the moisture tester had an average Moisture Content Reading Variation of $\pm 0.12\%$.

The results were then compared with the oven method. The average moisture content reading deviation was 2.10%. The minimum moisture content deviation was 0.15% at 22.3% MC_{wb}, oven method (0.68% error) while the maximum moisture content deviation was 3.91% at 16.4% MC_{wb} oven method (23.9% error). The meter was easy to use and light in weight.

PURPOSE AND SCOPE OF TEST

The Grain Moisture Tester was tested to establish its performance characteristics. The test was conducted from June 4 to 24, 2013 at the AMTEC Test Laboratory. It was evaluated based on its operating performance compared to the oven method of determining the moisture content of the palay.

OPERATOR'S MANUAL

The Operating Manual supplied with the meter contains the following: Instrument structure sketch, Preparation before use, Moisture measurement, Error correction, Calibration, How to reset factory calibration data, Status indication, Instruments additional functions, Accessories, Main technical index, Corresponding table of variety code, and Maintenance.

METHOD OF TEST

The INTELLIGENT LDS-1G Grain Moisture Tester was tested based on ISO 7700/1 (Check the Calibration of Moisture Meter, Part 1: Moisture meters for Cereals).

DESCRIPTION OF THE IMPLEMENT

The INTELLIGENT LDS-1G Grain Moisture Tester was a portable, digital, capacitance type of moisture tester (**Figure 1**). It consisted of the measuring section, battery compartment, main body, and control panel. The measuring section was where the sample was poured for moisture content measurement. The battery compartment houses four pieces of 1.5 volts, size AA, dry cell batteries which provided the power to the unit. The main body of the unit was provided with a handle grip. The control panel (**Figure 2**) consisted of a display window and four keys: the POWER Key to power on and off the unit; the SPECIES Key to select a crop code for moisture testing; the ARROWS (UP and DOWN) Keys to make selections; and, the OK Key to confirm the selection. The moisture content reading was read directly from the display window.

The moisture tester was provided with the following accessories (**Figure 3**): AC adapter, cleaning brush, blanking cylinder, funnel, and calibration weight.

Detailed specifications of the machine are shown in **Table 1**.



Figure 1. INTELLIGENT Grain Moisture Tester
Model: LDS-1G



Figure 2. Control Panel



Figure 3. INTELLIGENT LDS-1G
Grain Moisture Tester
and its accessories

MOISTURE METER SPECIFICATIONS

Table 1. Detailed specifications of the INTELLIGENT LDS-1G Grain Moisture Tester

1.	Brand	:	INTELLIGENT
2.	Model	:	LDS-1G
3.	Serial No.	:	20120713-2
4.	Manufacturer	:	China
5.	Dimensions and weight		
5.1	Overall length (mm)	:	198
5.2	Overall width (mm)	:	117
5.3	Overall height (mm)	:	198
5.4	Net Weight (g)	:	713.3
6.	Measuring principle	:	Capacitance
7.	Measuring error	:	$\leq \pm 0.5\%$
8.	Repetitive error	:	$\leq \pm 0.2\%$
9.	Measuring range (%)	:	3 - 35
10.	Operating Temperature ($^{\circ}\text{C}$)	:	0 - 40
11.	Display mode	:	High-brightness backlight LCD
12.	Power supply	:	Dry cells (1.5v, “AA” size x 4pcs) or external 9V DC stabilized power source

Table 1. Detailed specifications of the INTELLIGENT LDS-1G Grain Moisture Tester (Cont'n.)

13.	Applications	:	Foodstuff and other non-metal granular samples, such as rice grain, wheat, corn, soybean, rapeseed, etc.
14.	Accessories	:	AC adapter, cleaning brush, user's manual, blanking cylinder, funnel, calibration weight for each meter, four AA alkaline batteries

PERFORMANCE TEST

Results of test are shown in **Table 2**.

Moisture Tester on Test : INTELLIGENT
Model: LDS-1G

Crop used : Palay

Date of Test : June 4-24, 2013

Table 2. Results of performance test

INTELLIGENT MOISTURE TESTER, MODEL: LDS-1G		OVEN METHOD** (MC _{WB})	ERROR (%MC _{WB})	PERCENT ERROR
READING* (%MC _{WB})	VARIATION IN READING (%)			
23.8	±0.045	22.2	1.64	7.41
23.3	±0.110	23.1	0.19	0.82
22.1	±0.329	22.3	0.15	0.68
21.0	±0.055	21.5	0.54	2.52
20.3	±0.055	20.7	0.48	2.32
19.9	±0.207	20.8	0.96	4.60
16.6	±0.195	18.3	1.75	9.58
15.8	±0.259	18.1	2.32	12.8
14.6	±0.045	18.1	3.44	19.0
13.5	±0.207	16.5	2.94	17.8
12.5	0.055	16.4	3.91	23.9
12.0	±0.122	8.29	3.71	44.8
11.2	±0.055	7.69	3.55	46.1
10.4	±0.071	6.97	3.43	49.3

* Average of five samples

** Average of three samples

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N.B.

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